

Anant Jain

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EDUCATION

BITS Pilani, K.K. Birla Goa Campus

Bachelor of Engineering in Electronics and Instrumentation

Goa, India

- CGPA: 6.61

EXPERIENCE

Bhaskaracharya National Institute for Space Applications and Geo-Informatics (BISAG-N)

May 2026 – July 2026

Summer Intern — Computer Vision and Deep Learning

Gandhinagar, Gujarat

- Developed **NeuralBoard**, a YOLOv11-based PCB component detection and classification system identifying 9 component classes from real PCB photographs.
- Built a multi-dataset training pipeline across 3 annotation formats (VIA CSV/JSON, Pascal VOC XML, YOLO TXT) and 3 datasets, improving detector mAP50 from 0.162 to 0.460 over 10+ training runs.
- Trained a YOLOv11 classifier on 37,023 images achieving **99.49% Top-1 accuracy**; built an enhancement pipeline (LANCZOS4 4x + CLAHE + sharpen) improving resistor detection by 200%.
- Implemented PCB image validation, interactive component filtering, GPU training on RTX 4050, and ONNX model export for deployment.

PROJECTS

NeuralBoard — PCB Component Detection & Classification | *YOLOv11, PyTorch, OpenCV, Python*

- End-to-end CV pipeline for automated PCB inspection using YOLOv11 detection and classification trained on 37k+ annotated images. Achieved mAP50 of 0.931 on diodes and 0.774 on transistors.
- Includes PCB validation (HSV + edge density), 4x LANCZOS4 enhancement, interactive component filtering, and ONNX export. Paired with a dark-themed web frontend with live BOM panel and CSV/JSON export.

Multi-Dataset Annotation Conversion Pipeline | *Python, xml.etree.ElementTree*

- Scripts to parse and convert VIA CSV (embedded JSON), Pascal VOC XML, and YOLO TXT annotation formats into a unified YOLO training dataset, handling Windows MAX_PATH limits, encoding issues, and class ID remapping.

NeuralBoard Web Frontend | *HTML5, CSS3, JavaScript*

- Dark-themed SPA with login, camera (WebRTC) and file upload, live Bill of Materials table, CSV/JSON export, shopping cart, and scan history.

TECHNICAL SKILLS

Languages: Python (Advanced), C (Basic), HTML/CSS/JavaScript

Deep Learning & CV: YOLOv11 (Ultralytics), PyTorch 2.5.1+cu121, OpenCV, NumPy, Pillow

ML Concepts: Object Detection, Image Classification, Transfer Learning, Data Augmentation, mAP50/Precision/Recall

Tools: VS Code, Roboflow, CUDA/GPU Training (NVIDIA RTX 4050), ONNX Export

Interests: Computer Vision, Deep Learning, Core Electronics, Embedded Systems

OTHER INFORMATION

Strengths: Quick learner, eager to gain hands-on experience and develop technical skills

Sports: Cricket, Badminton, Volleyball

Co-Curricular: Attended technical sessions and workshops; presented NeuralBoard at BISAG-N final seminar

Objective: To gain practical industry exposure and apply theoretical knowledge while developing professional skills